EE2001 Electrical Circuits- I:

Continuation to:

Response of First-Order RL and RC Circuits

5 December 2023

Outline



Continuation to:

3. **Natural Response of an RC Circuit**
4. **The Step and the Total Responses of RL and RC Circuits**

The Natural Response of an RC Circuit

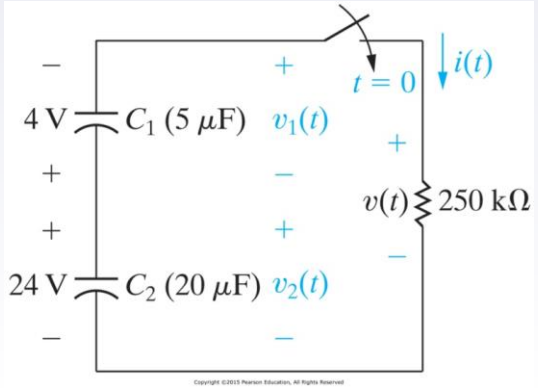
- Example-4: The initial voltages on capacitors C_1 and C_2in the given circuit have been established by sources not shown. The switch is closed at $t = 0$.

a) Find $v_1(t)$, $v_2(t)$ and $v(t)$ for $t \geq 0$ and $i(t)$

b) Calculate the initial energy stored in the capacitors C_1 and C_2.

c) Determine how much energy is stored in the capacitors as $t \to \infty$

d) Show that the total energy delivered to the 250 kΩ resistor is the difference between the results obtained in (b) and (c).



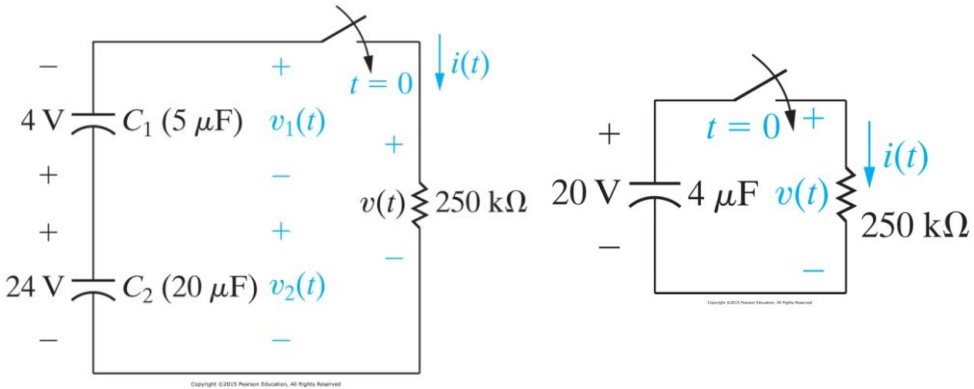
a) Once we know $v(t)$ we can obtain $i(t)$ from Ohm's law. After determining $i(t)$, we can calculate $v_1(t)$ and $v_2(t)$ because the voltage across a capacitor is a function of the capacitor current. To find $v(t)$, we replace the series-connected capacitors with an equivalent capacitor with a capacitance of 4 μF and is charged to a voltage of 20 V.

The Natural Response of an RC Circuit

- Example-4: The initial voltages on capacitors C_1 and C_2in the given circuit have been established by sources not shown. The switch is closed at $t = 0$.

a) Find $v_1(t)$, $v_2(t)$ and $v(t)$ for $t \geq 0$ and $i(t)$

b) Calculate the initial energy stored in the capacitors C_1 and C_2.



a) Therefore the circuit reduces to a simpler topology, which reveals that the initial value of $v(t)$ is 20 V, and that the time constant of the circuit is $(4)(250) \times 10^{-3}$, or 1 s. Thus the expression for $v(t)$ is,

$$v(t) = 20e^{-t}\ V, \qquad t \geq 0$$

The current $i(t)$ is,

$$i(t) = \frac{v(t)}{250000} = 80e^{-t}\ \mu A, \qquad t \geq 0^+$$

Using $i(t)$, we can calculate the expressions for $v_1(t)$ and $v_2(t)$:

$$v_1(t) = -\frac{10^6}{5}\int_0^t 80 \times 10^{-6}e^{-x}dx - 4$$

$$= (16e^{-t} - 20)\ V, \qquad t \geq 0$$

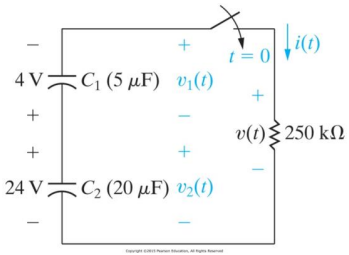
The Natural Response of an RC Circuit

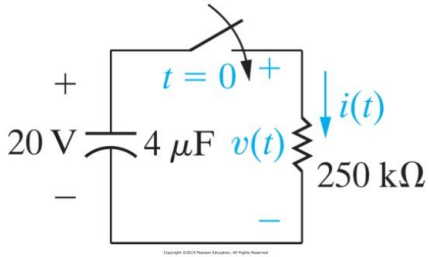
- Example-4: The initial voltages on capacitors C_1 and C_2in the given circuit have been established by sources not shown. The switch is closed at $t = 0$.

a) Find $v_1(t)$, $v_2(t)$ and $v(t)$ for $t \geq 0$ and $i(t)$

b) Calculate the initial energy stored in the capacitors C_1 and C_2.

c) Determine how much energy is stored in the capacitors as $t \to \infty$





$$v_2(t) = -\frac{10^6}{20}\int_0^t 80 \times 10^{-6} e^{-x} dx + 24$$

$$= (4e^{-t} + 20)\ V, \qquad t \geq 0$$

b) The initial energy stored in C_1 is

$$w_1 = \frac{1}{2}(5 \times 10^{-6})(16) = 40\ \mu J$$

The initial energy stored in C_2 is

$$w_2 = \frac{1}{2}(20 \times 10^{-6})(576) = 5760\ \mu J$$

The total energy stored in the two capacitors

$$w_o = 40 + 5760 = 5800\ \mu J$$

c) As $t \to \infty$, $v_1 \to -20\ V$ and $v_2 \to +20\ V$

Therefore the energy stored in the capacitors is

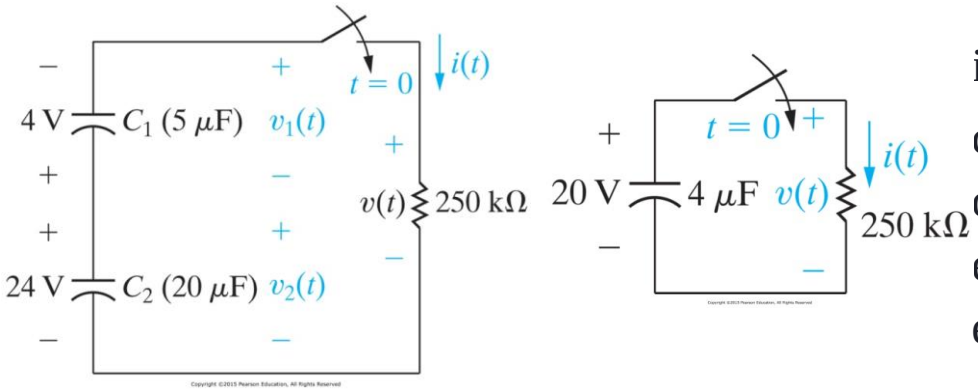
$$w_\infty = \frac{1}{2}(5 + 20) \times 10^{-6}(400) = 5000\ \mu J$$

The Natural Response of an RC Circuit

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- Example-4: The initial voltages on capacitors C_1 and C_2in the given circuit have been established by sources not shown. The switch is closed at $t = 0$.

d) Show that the total energy delivered to the 250 kΩ resistor is the difference between the results obtained in (b) and (c).



d) The total energy delivered to the 250 k Ω resistor is

$$w = \int_0^\infty p dt = \int_0^\infty \frac{400e^{-2t}}{250000} dt = 800\ \mu J$$

Comparing the results obtained in (b) and (c) shows that

$$800\ \mu J = (5800 - 5000)\ \mu J$$

The energy stored in the equivalent capacitor is $\frac{1}{2}(4 \times 10^{-6})(400)$, or 800 μJ. Because this capacitor predicts the terminal behavior of the original series-connected capacitors, the energy stored in the equivalent capacitor is the energy delivered to the 250 kΩ resistor.

The Step and the Total Responses of RL and RC Circuits

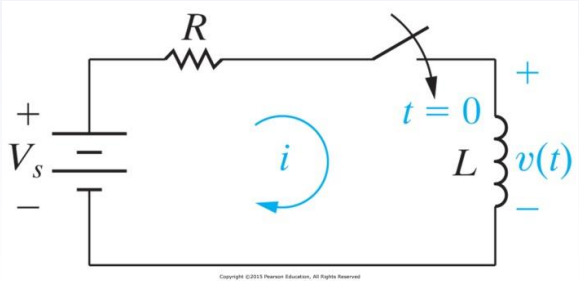
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- The response of a circuit/system to a sudden application of an external constant excitation (i.e., a constant current or a constant voltage source) is referred to as the step response of the circuit/system.
- For the step response, the initial conditions in the circuit/system are assumed to be zero.
- The response of a circuit/system to a sudden application of an external excitation when there are also non-zero initial conditions is referred to as the total response of the circuit/system.
- Therefore, the total response is the natural response and the step response (or the response to another input) together.

The Step and the Total Responses of RL and RC Circuits

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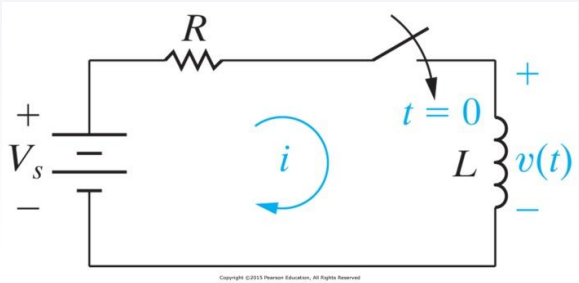
- Consider the circuit



- If $i(0) = 0$, step response.
- If $i(0) = I_0 \neq 0$, total response.
- We can write a loop (KVL) equation to solve for the response: $-V_s + Ri + L\frac{di}{dt} = 0$

The Step and the Total Responses of RL and RC Circuits

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- If $i(0) = 0$ (step response)

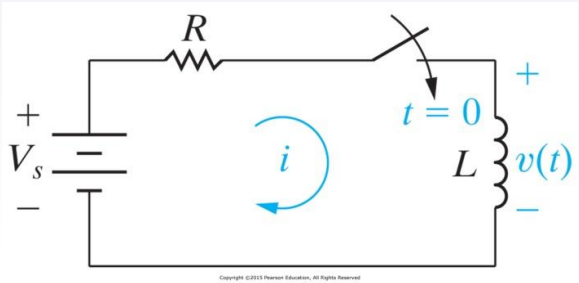
$$i(t) = \frac{V_s}{R} - \frac{V_s}{R} e^{-\frac{R}{L}t}$$

- If $i(0) = I_0 \neq 0$ (total response)

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right) e^{-\frac{R}{L}t}$$

The Step and the Total Responses of RL and RC Circuits

- Consider the total response



$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-\frac{R}{L}t}$$

$$v(t) = (V_s - I_0 R)e^{-\frac{R}{L}t}$$

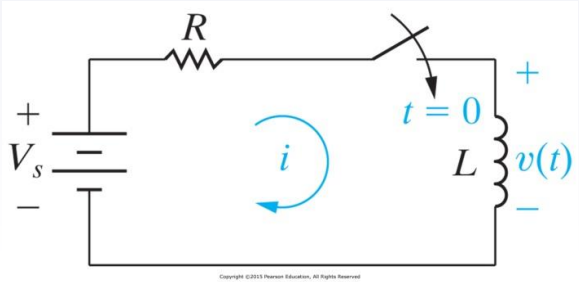
- It can be rewritten in the form

$$i(t) = \underbrace{I_0 e^{-\frac{R}{L}t}}_{\text{natural response}} + \underbrace{\frac{V_s}{R}\left(1 - e^{-\frac{R}{L}t}\right)}_{\text{step response}}$$

- The first part is the response due to the initial conditions (natural response)
- The second part is the response due to the input (step response)

The Step and the Total Responses of RL and RC Circuits

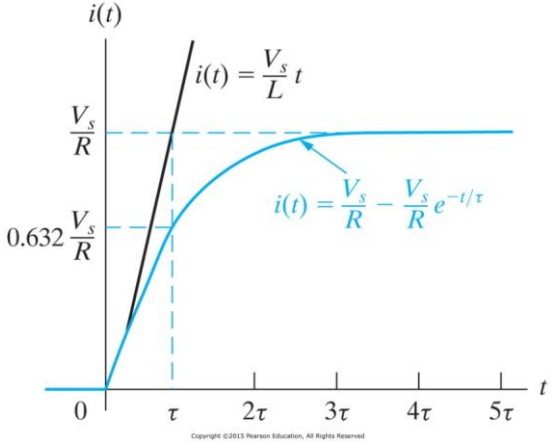
- Consider the step response



$$i(t) = \frac{V_s}{R} - \frac{V_s}{R} e^{-\frac{R}{L}t}$$

$$v(t) = V_s e^{-\frac{R}{L}t}$$

- The plot of the response is in the form



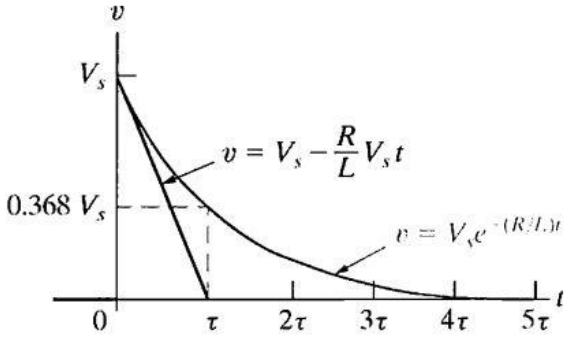


Figure 7.18 ▲ Inductor voltage versus time.

End of the Lesson (05.12.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist